

The Implication Lattice of Real Analysis

Limits, continuity, differentiability, integrability, and convergence

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Abstract

The relations among having a limit, continuity, differentiability, Riemann integrability, and convergence are the substance of a first course in analysis, but they are established piecemeal, each where it is needed, and are nowhere assembled. We assemble them. For every ordered pair of these properties we record either an implication, with proof, or a non-implication, with an explicit function that witnesses it. Four of the five properties are local and one is not, so the material divides into three parts: the local implications at a point, the global ones on a compact interval together with integrability, and the question of which properties survive passage to a limit. Lebesgue's criterion makes the integrability row exact rather than illustrative, and Volterra's example shows that the two halves of the Fundamental Theorem of Calculus are independent statements. Seventeen implications and twelve counterexamples are recorded.

1 Introduction

Five properties recur throughout elementary real analysis: a function may have a limit at a point, be continuous there, be differentiable there, be Riemann integrable on an interval, and a sequence of functions may converge. Every course establishes relations among them — differentiability implies continuity, continuity on a compact interval implies integrability, uniform convergence preserves continuity — but each relation is proved where it is needed, and the resulting map is never drawn. A student who has finished Rudin [2] knows all the arrows individually and has rarely seen them together.

This note draws the map. For each ordered pair of properties we record a verdict: an implication with a proof, or a non-implication with a specific function. Nothing is asserted without one or the other.

A structural remark

Four of the five properties are *local*: they are properties of a function at a point. Riemann integrability is not. It is a property of a function on an interval, and to ask whether f is integrable *at* a point is meaningless. A single five-by-five table would silently equivocate between the two kinds, so the material divides into three parts.

- **The local theory** (§2): the properties of f at a single point p .
- **The global theory** (§3): those properties holding at every point of $[a, b]$, together with integrability.

- **The passage to the limit** (§4): given $f_n \rightarrow f$, which properties of the f_n are inherited by f .

The passage to the limit is not a separate subject. It is the local and global theories asked a second time with a limit in front, and the answers change — which is precisely the reason uniform convergence was isolated as a notion.

Conventions and sources

Numbered references of the form *Rudin 4.19* are to [2]. Two lemmas on the oscillation of a function are used in §3 without proof; they are proved in [1], whose appendices develop the oscillation machinery in detail, and also appear in [5]. Where a comparison with another standard treatment is illuminating we cite Tao [3, 4], the Tsinghua lectures of Yu Pin [5], or the East China Normal University text [6].

Counterexamples are labelled E1–E12 and collected with full dossiers in §5; they are referred to by label throughout.

1.1 The five properties

We fix the definitions used throughout. Everything takes place in $\mathbb{R}$; E denotes a subset of $\mathbb{R}$ and f a real-valued function on E . Recall that p is a *limit point* of E if every neighbourhood of p contains a point of E other than p .

Definition 1.1 (Limit). Let p be a limit point of E . We write $\lim_{x \rightarrow p} f(x) = q$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - q| < \varepsilon \quad \text{for all } x \in E \text{ with } 0 < |x - p| < \delta.$$

Definition 1.2 (Continuity). Let $p \in E$. Then f is *continuous at p* if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(p)| < \varepsilon \quad \text{for all } x \in E \text{ with } |x - p| < \delta,$$

and *continuous on E* if it is continuous at every point of E .

The two definitions differ in exactly two places, and both differences matter. Definition 1.1 excludes $x = p$ and does not mention $f(p)$ — the limit is indifferent to the value, and need not be defined at p at all. Definition 1.2 includes $x = p$ and names $f(p)$ as the target. Figure 1 draws the difference.

Definition 1.3 (Differentiability). Let f be defined on an interval I and let $p \in I$. Then f is *differentiable at p* if

$$f'(p) = \lim_{x \rightarrow p} \frac{f(x) - f(p)}{x - p}$$

exists as a finite limit, in the sense of Definition 1.1 applied to the difference quotient on $I \setminus \{p\}$.

Definition 1.4 (Riemann integrability). Let f be bounded on $[a, b]$. A *partition* P is a finite set $a = x_0 < x_1 < \cdots < x_n = b$; write $\Delta x_i = x_i - x_{i-1}$ and

$$M_i = \sup_{[x_{i-1}, x_i]} f, \quad m_i = \inf_{[x_{i-1}, x_i]} f, \quad U(P, f) = \sum_i M_i \Delta x_i, \quad L(P, f) = \sum_i m_i \Delta x_i.$$

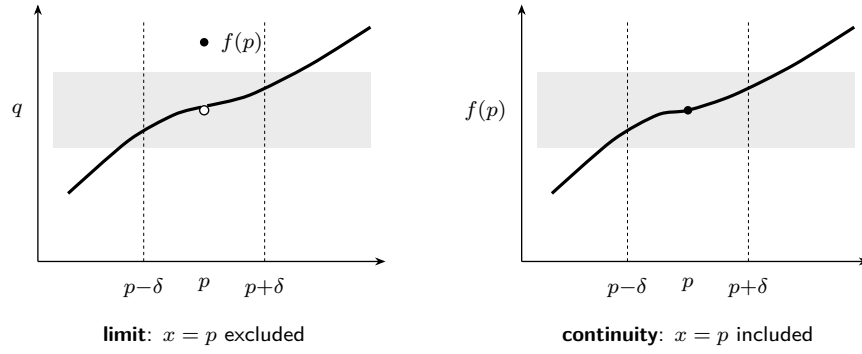


Figure 1: The two definitions side by side. In both, whatever enters the vertical δ -strip must leave through the horizontal ε -band. For the limit the point $x = p$ is punched out of the strip, so $f(p)$ — drawn far from the band on the left — is irrelevant and may even be undefined. For continuity the point is included and the band is centred on $f(p)$ itself.

The *upper* and *lower integrals* are $\overline{\int_a^b} f = \inf_P U(P, f)$ and $\underline{\int_a^b} f = \sup_P L(P, f)$; these always exist because f is bounded. If they are equal, f is *Riemann integrable* on $[a, b]$ and their common value is $\int_a^b f$.

Lemma 1.5 (Riemann's criterion). *A bounded f is integrable on $[a, b]$ if and only if for every $\varepsilon > 0$ there is a partition P with $U(P, f) - L(P, f) < \varepsilon$.*

Proof. Refining a partition neither raises U nor lowers L , so every lower sum is at most every upper sum and $\underline{\int} f \leq \overline{\int} f$. If some P has $U(P, f) - L(P, f) < \varepsilon$ then the two integrals differ by less than ε , and as ε was arbitrary they agree. Conversely if they agree, choose P_1 with $U(P_1, f)$ within $\varepsilon/2$ of the common value and P_2 with $L(P_2, f)$ within $\varepsilon/2$ of it; their common refinement works. $\square$

Definition 1.6 (Convergence). Let f_n and f be defined on E . The sequence (f_n) converges to f *pointwise* if $f_n(x) \rightarrow f(x)$ for every $x \in E$, and *uniformly* if

$$\sup_{x \in E} |f_n(x) - f(x)| \rightarrow 0,$$

equivalently if for every $\varepsilon > 0$ there is an N , *depending on ε alone*, with $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ and all $x \in E$ simultaneously.

Uniform convergence implies pointwise convergence and not conversely; the whole of §4 is an examination of the gap between them.

The lattice

Three features of Figure 2 are worth stating explicitly. The top row is a strict chain, and each strictness is witnessed by one explicit function. The bottom row is not the top row repeated: integrability is genuinely global, and the arrow into it comes from compactness rather than from anything pointwise. And there is no arrow back out of integrability at all.

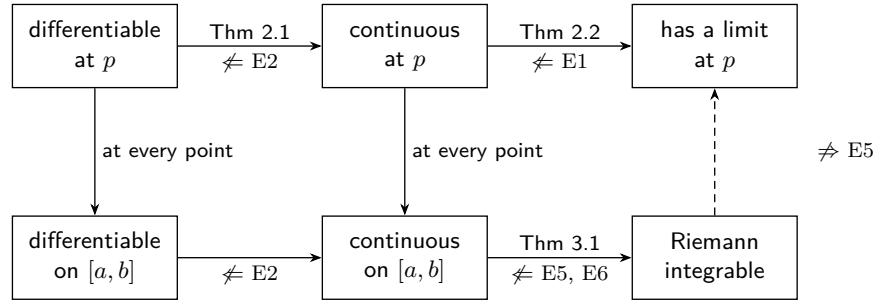


Figure 2: The local and global implications. Solid arrows are theorems; the label beneath each records the failure of its converse and names the function that witnesses it. Integrability implies nothing pointwise: an integrable function may lack a two-sided limit at a jump (E5) and may be discontinuous on a dense set (E6); conversely, boundedness alone is not enough for integrability (E7).

2 Local implications

Throughout this section p is a limit point of the domain of f .

Theorem 2.1. *If f is differentiable at p then f is continuous at p .*

Proof. For $x \neq p$,

$$f(x) - f(p) = \frac{f(x) - f(p)}{x - p} \cdot (x - p).$$

As $x \rightarrow p$ the first factor tends to $f'(p)$ and the second to 0, so the product tends to 0. $\square$

The proof uses the existence of $f'(p)$ only to know that the difference quotient has some finite limit. It therefore proves slightly more: if the difference quotient is bounded near p — in particular if f satisfies a Lipschitz condition at p — then f is continuous at p .

Theorem 2.2. *If f is continuous at p then $\lim_{x \rightarrow p} f(x)$ exists and equals $f(p)$.*

Proof. Immediate from the definitions, p being a limit point of the domain. $\square$

Remark 2.3. The hypothesis that p is a limit point cannot be dropped. At an isolated point of the domain every function is continuous, while $\lim_{x \rightarrow p} f(x)$ is undefined, since the usual definition requires p to be approached through points of the domain other than p itself. This is the distinction between deleted and undeleted limits; Tao [3, §9.3] adopts the undeleted convention and obtains a correspondingly different statement of Theorem 2.2.

Both converses fail, and cheaply: E1 has a limit at 0 without being continuous there, and $E2 = |x|$ is continuous at 0 without being differentiable there. Since $|x|$ is Lipschitz, the hypothesis of the strengthening noted after Theorem 2.1 is genuinely weaker than differentiability.

Theorem 2.4 (Darboux). *If f is differentiable on $[a, b]$ then f' assumes every value between $f'(a)$ and $f'(b)$, whether or not f' is continuous.*

How the proof runs

- (1) *Locate the obstacle.* f' is not assumed continuous, so the intermediate value theorem cannot be applied to it. Some other mechanism must produce a point where f' takes a prescribed value.
- (2) *Find the one available mechanism.* A differentiable function attains an interior extremum only where its derivative vanishes. That produces the value 0, and only the value 0.
- (3) *Move the target to 0.* Subtract the linear function of slope λ . The statement “ f' takes the value λ ” becomes “ g' takes the value 0”, which step (2) can deliver.
- (4) *Manufacture the interior extremum.* The signs $g'(a) < 0 < g'(b)$ force g to decrease at the left end and increase at the right, so its minimum over $[a, b]$ cannot be at either endpoint. Compactness supplies the minimum; step (2) reads off $g' = 0$ there.
- (5) *Audit.* Compactness is spent once, in step (4), to know the minimum exists. Differentiability is spent twice: at the endpoints, for the signs, and at the interior extremum. Continuity of f' is never used — which is the whole point.

Figure 3 draws the dependencies among these steps, and Figure 4 the geometry of the statement.

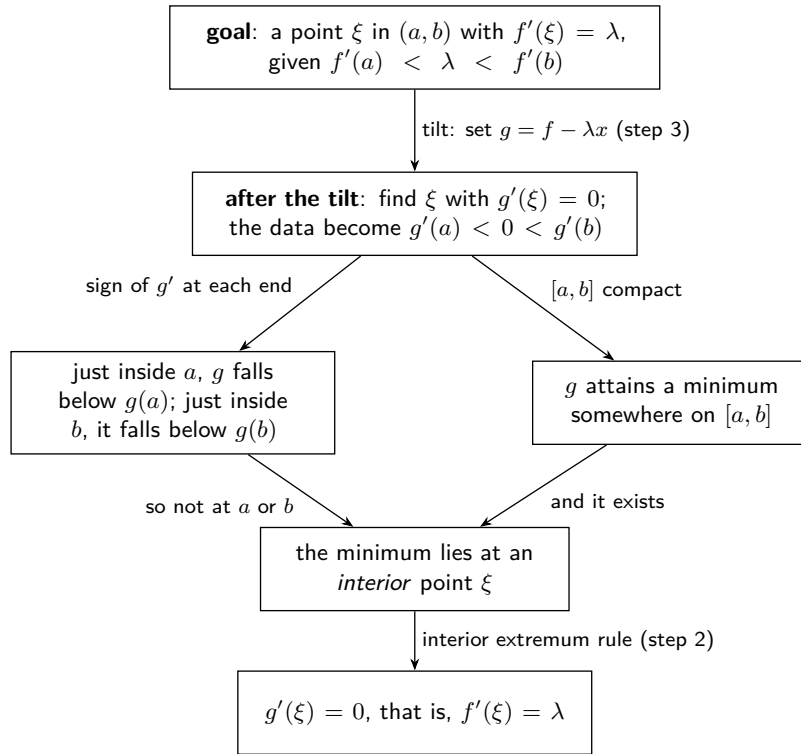


Figure 3: The Darboux proof as a flow chart: each box is a statement, and each arrow carries the reason the next statement follows. The tilt trades the target value λ for 0. After it, two independent observations meet in the middle — the signs of g' push g below both endpoint values, and compactness makes the minimum exist at all — so the minimum exists and is not at an endpoint. At an interior minimum the derivative must vanish, and untilting turns $g'(\xi) = 0$ back into $f'(\xi) = \lambda$. The step numbers refer to the walk above.

Proof. Suppose $f'(a) < \lambda < f'(b)$ and set $g(x) = f(x) - \lambda x$, so that $g'(a) < 0 < g'(b)$. From $g'(a) < 0$ we get $g(t) < g(a)$ for some t just to the right of a , and from $g'(b) > 0$ we get $g(s) < g(b)$ for some s just to the left of b . Hence the minimum of g over the compact interval $[a, b]$ is attained at an interior point ξ , where $g'(\xi) = 0$; that is, $f'(\xi) = \lambda$. $\square$

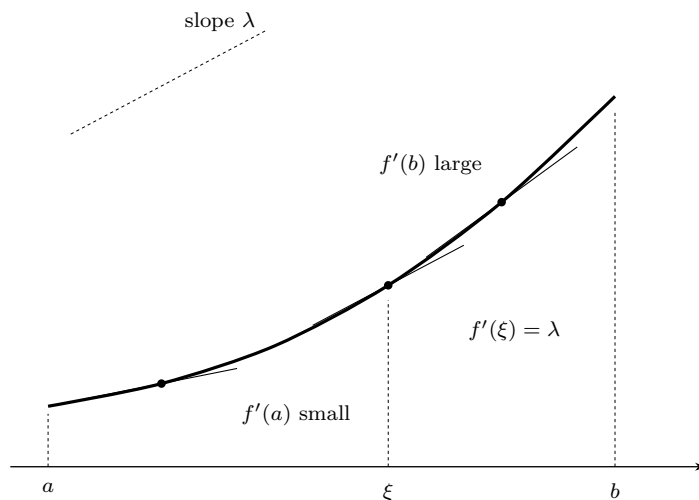


Figure 4: Darboux's theorem. The tangent slope varies from $f'(a)$ to $f'(b)$ along the curve and must pass through every intermediate value λ , even though f' need not vary continuously. The proof locates ξ not by following the slope but by tilting the graph until the target slope becomes horizontal, so that an interior extremum can be produced.

Corollary 2.5. *A derivative has no simple discontinuities. In particular the signum function is not the derivative of any function.*

Proof. At a simple discontinuity x both one-sided limits of f' exist, and since f' is discontinuous at x , at least one of them — say the right-hand limit L — differs from $f'(x)$. Pick λ strictly between $f'(x)$ and L . Because $f'(t) \rightarrow L$ as $t \downarrow x$, there is an interval $(x, x + \delta)$ on which f' stays near L and therefore omits λ . But for each t in that interval, Theorem 2.4 applied to $[x, t]$ forces f' to attain every value between $f'(x)$ and $f'(t)$ — in particular λ — at some point of (x, t) . Contradiction. $\square$

Corollary 2.5 says a derivative may fail to be continuous only by oscillating or by becoming unbounded, never by jumping. That the permitted behaviour occurs is shown by E3: the function $x^2 \sin(1/x)$ is differentiable everywhere and its derivative oscillates without limit at the origin (Figure 5). Consequently *differentiable* and *continuously differentiable* are distinct hypotheses, which is why the class $\mathcal{C}^1$ requires a separate name.

2.1 How wide is the gap?

The chain of Theorems 2.1 and 2.2 is strict, but saying so does not say *what is missing*. This subsection answers that: for each step, exactly what must be added.

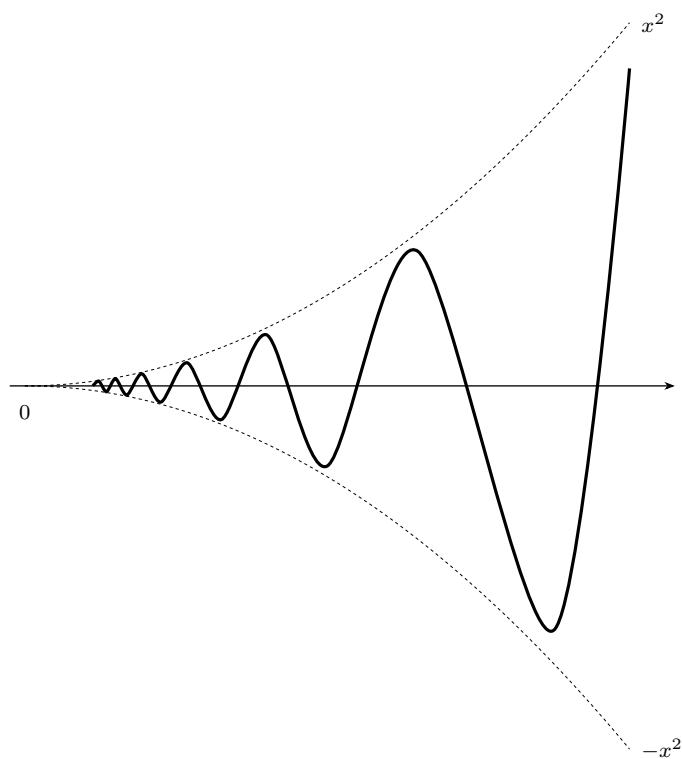


Figure 5: E3, the function $x^2 \sin(1/x)$, trapped between the envelopes $\pm x^2$. The envelope closes fast enough to force $f'(0) = 0$, while the oscillation quickens fast enough that f' has no limit at the origin: differentiability at a point says nothing about the behaviour of the derivative near it.

From a limit to continuity: one number

Proposition 2.6. *Let $p \in E$ be a limit point of E . Then $\lim_{x \rightarrow p} f(x)$ exists if and only if there is a function g on E with $g = f$ on $E \setminus \{p\}$ and g continuous at p ; and then $g(p) = \lim_{x \rightarrow p} f(x)$.*

Proof. If the limit is q , set $g(p) = q$ and $g = f$ elsewhere; the ε - δ condition of Definition 1.1 becomes that of Definition 1.2 because the value at p now sits inside every band. Conversely, if such a g exists, then for $x \neq p$ we have $f(x) = g(x)$, so $f(x) \rightarrow g(p)$. $\square$

So the deficit at the first step is a single number: having a limit means being continuous *up to one value*, and E1 is what it looks like when that value is set wrongly. Nothing about f away from p needs to change.

From continuity to differentiability: one order

The second step is not one number but one *order of approximation*.

Theorem 2.7. *f is differentiable at p with $f'(p) = A$ if and only if*

$$f(x) = f(p) + A(x - p) + o(|x - p|) \quad (x \rightarrow p).$$

Proof. For $x \neq p$, divide the displayed identity by $x - p$: it says exactly that $(f(x) - f(p))/(x - p) - A \rightarrow 0$, which is Definition 1.3. $\square$

Now place the two properties side by side. Continuity at p says

$$f(x) = f(p) + o(1),$$

that is, f is approximable near p by a *constant*, with error tending to 0. Differentiability says f is approximable by an *affine* function, with error tending to 0 *faster than* $|x - p|$. The gap between them is one order of approximation — and, geometrically, the difference between a horizontal band and a wedge (Figure 6).

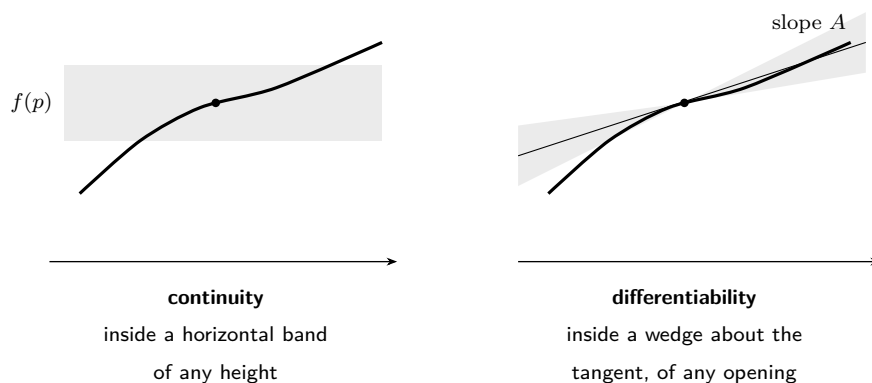


Figure 6: One order of approximation. Continuity requires the graph to enter every horizontal band about $f(p)$; differentiability requires it to enter every *wedge* about the line of slope A — a band whose height shrinks in proportion to $|x - p|$. The second condition is strictly stronger because the target narrows as the point is approached.

The rungs in between

The gap is not a single step. For $0 < \alpha \leq 1$ call f α -Hölder at p if $|f(x) - f(p)| \leq C|x - p|^\alpha$ for some constant C and all x near p ; the case $\alpha = 1$ is the Lipschitz condition. Then

$$\text{differentiable} \implies \text{Lipschitz} \implies \alpha\text{-Hölder } (\alpha < 1) \implies \text{continuous},$$

and every implication is strict, with an explicit witness at each rung:

witness	has	lacks
$ x $	Lipschitz at 0	differentiable at 0
$\sqrt{ x }$	$\frac{1}{2}$ -Hölder at 0	Lipschitz at 0
$1/\log(1/ x)$	continuous at 0	α -Hölder for any $\alpha > 0$

The last needs a word: with $f(0) = 0$ the quotient $|f(x)|/|x|^\alpha = 1/(|x|^\alpha \log(1/|x|))$ tends to ∞ for every $\alpha > 0$, since a power beats a logarithm. So no Hölder condition whatever holds, and continuity really is the bottom rung (Figure 7).

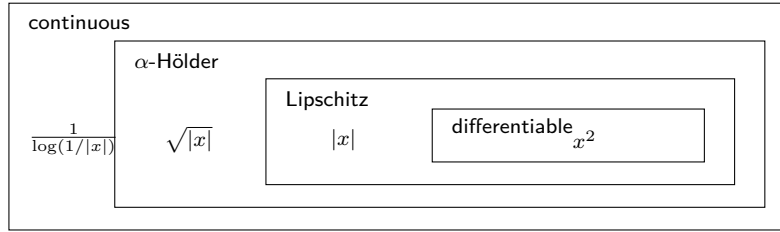


Figure 7: The rungs between continuity and differentiability, with the function that sits in each annulus. Differentiability is a good deal more than continuity: three strict inclusions separate them, and by Theorem 2.8 the innermost box is not merely smaller but negligible.

The gap is enormous

Theorem 2.8. *The set of $f \in \mathcal{C}[a, b]$ that are differentiable at even one point is of first category. Consequently the generic continuous function is differentiable nowhere.*

This is the Baire-category statement behind E8; see [5, Lect. 28]. It settles the question of size: passing from continuity to differentiability is not a technical refinement but a restriction to a negligible subclass. Weierstrass's function is not a curiosity — the differentiable functions are.

What buys the gap back

Three standard hypotheses close it, each at a price.

- *Finiteness and agreement of the four Dini derivatives.* With D^+, D_+, D^-, D_- the upper and lower right- and left-hand limits of the difference quotient, f is differentiable at p exactly when all four are finite and equal. This is a restatement rather than a repair, but it isolates the two separate deficits: *finiteness* is the Lipschitz-type bound, and *agreement*

is the choice of a single direction. For $|x|$ at 0 all four are finite — $+1$ on the right, -1 on the left — and they fail only to agree.

- *Monotonicity.* A monotone function on $[a, b]$ is differentiable almost everywhere (Lebesgue). The price is that “everywhere” weakens to “almost everywhere”: monotonicity cannot rule out a single corner, only a large set of them.
- *A Lipschitz condition.* By Rademacher’s theorem a Lipschitz function is differentiable almost everywhere — again only almost everywhere, as $|x|$ shows.

The pattern is worth naming. Nothing short of differentiability buys differentiability *everywhere*; the natural hypotheses buy it only off a null set. That is the same trade Lebesgue’s criterion makes in §3, and it is why “almost everywhere” becomes the working notion from Chapter 11 of [2] onwards.

Proposition 2.9. *If $\lim_{x \rightarrow p} f(x)$ exists then f is bounded on some punctured neighbourhood of p .*

Proof. Take $\varepsilon = 1$ in the definition of the limit: there is a $\delta > 0$ with $|f(x) - q| < 1$, hence $|f(x)| < |q| + 1$, throughout the punctured δ -neighbourhood. $\square$

The chain therefore extends one step downward,

$$\text{differentiable} \implies \text{continuous} \implies \text{has a limit} \implies \text{locally bounded},$$

with all three implications strict; the last is strict because $E4 = \sin(1/x)$ is bounded and has no limit at 0.

3 Integrability

Promoting each local property to “at every point of $[a, b]$ ” brings in a fourth property with no pointwise form. The arrow into it comes from compactness.

Theorem 3.1. *If f is continuous on $[a, b]$ then f is Riemann integrable there.*

How the proof runs

- (1) *Translate the goal.* Integrability asks that $U - L$ be small. By Figure 9, $U - L$ is the total area of the strips between sup and inf over the pieces of a partition. So the goal is: make every strip short at once.
- (2) *Identify what would suffice.* If every subinterval of a partition had oscillation below η , the total area would be at most $\eta(b - a)$. So it suffices to find a mesh size on which the oscillation is everywhere below η .
- (3) *Notice what continuity alone does not give.* Continuity supplies, at each point, a δ that works *there*. Those δ ’s could shrink to nothing as the point moves, leaving no single mesh that works everywhere.
- (4) *Spend compactness.* On a compact interval, continuity is uniform: one δ serves every point. This is the only step that uses anything about $[a, b]$ beyond its length.

- (5) *Reverse-engineer the constant.* Working back from the target $U - L \leq \varepsilon$ and the factor $\sum \Delta x_i = b - a$, the oscillation must be held below $\varepsilon/(b - a)$; that is the η to demand in step (2).
- (6) *Audit.* Compactness once, in step (4). Nothing pointwise beyond continuity, and nothing about differentiability, enters anywhere.

Figure 8 draws the dependencies among these steps.

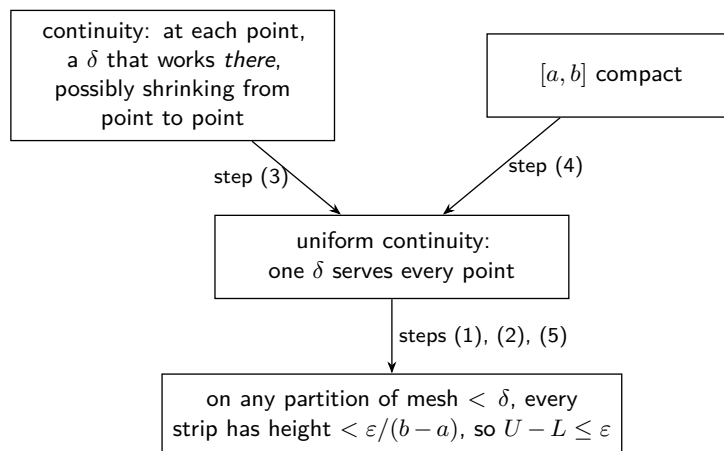


Figure 8: The proof of Theorem 3.1 as a flow chart. Continuity alone supplies only a pointwise δ ; compactness upgrades it to a single δ valid everywhere, and the rest is bookkeeping — a partition finer than that δ holds every oscillation below $\varepsilon/(b - a)$. The step numbers refer to the walk above.

Proof. Let $\varepsilon > 0$. Since $[a, b]$ is compact, f is uniformly continuous, so there is a $\delta > 0$ with $|f(x) - f(y)| < \varepsilon/(b - a)$ whenever $|x - y| < \delta$. Let P be any partition of mesh less than δ . On each subinterval the supremum M_i and infimum m_i of f differ by at most $\varepsilon/(b - a)$, whence

$$U(P, f) - L(P, f) = \sum_i (M_i - m_i) \Delta x_i \leq \varepsilon.$$

Riemann's criterion gives integrability. □

The only substantial step is the passage from continuity to *uniform* continuity, and it is the only place compactness is used; the rest is bookkeeping. In particular nothing pointwise beyond continuity enters, and differentiability plays no role.

3.1 Lebesgue's criterion

The examples below could each be settled by hand with upper and lower sums, which would leave the impression that integrability is a matter of luck. It is not.

Definition 3.2 (Measure zero). A set $E \subset \mathbb{R}$ has *measure zero* if for every $\varepsilon > 0$ there are countably many intervals $I_1, I_2, \dots$ with $E \subset \bigcup_k I_k$ and $\sum_k |I_k| < \varepsilon$.

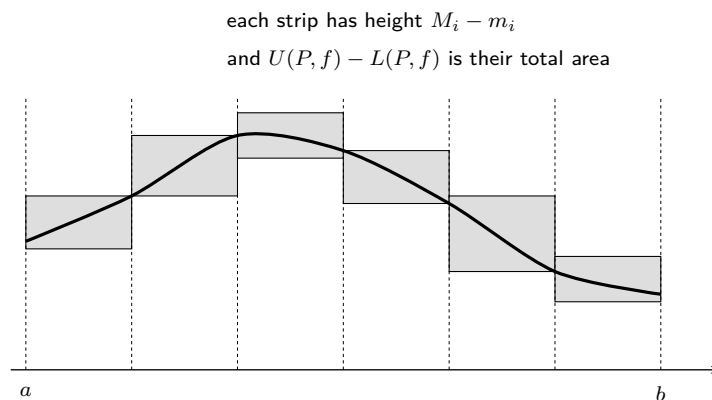


Figure 9: Upper and lower sums. The shaded strips are the oscillation of f over each subinterval, and their total area is exactly $U(P, f) - L(P, f)$. Uniform continuity bounds every strip's height by the same $\varepsilon/(b - a)$ once the mesh is small enough, so the total area is at most ε regardless of how the partition is placed.

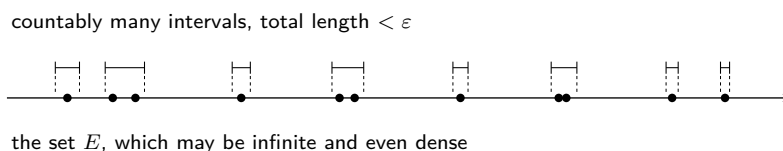


Figure 10: Measure zero. The intervals may be as numerous as one likes provided their lengths sum to less than ε , and ε is arbitrary. Every countable set qualifies, its k th point being covered by an interval of length $\varepsilon 2^{-k-1}$; no interval of positive length does.

The picture to keep in mind is Figure 10.
For a bounded f write

$$\omega_f(x) = \lim_{r \rightarrow 0^+} \left[\sup_{|y-x| < r} f(y) - \inf_{|y-x| < r} f(y) \right]$$

for the *oscillation* of f at x . We use two elementary facts, proved in [1, appendices to Chapters 2 and 4] and also in [5, Lect. 21]: f is continuous at x if and only if $\omega_f(x) = 0$; and for each $\varepsilon > 0$ the set $\Omega_\varepsilon = \{x : \omega_f(x) \geq \varepsilon\}$ is closed, so that the set D of discontinuities of f is $D = \bigcup_{n \geq 1} \Omega_{1/n}$.

Theorem 3.3 (Lebesgue's criterion). *A bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero.*

How the proof runs

- (1) *Fix the currency.* Both hypotheses are statements that something is small: integrability says an *area* is small, measure zero says a *length* is small. The proof is an exchange rate between the two, and oscillation is the exchange rate — $\text{area} \approx \text{oscillation} \times \text{length}$.
- (2) *Forwards: read the area as a length.* A subinterval whose interior meets $\Omega_{1/n}$ has oscillation at least $1/n$, so it contributes at least $(1/n)\Delta x_i$ to $U - L$. If $U - L < \eta/n$, those subintervals have total length below η — and that is literally a covering of $\Omega_{1/n}$ by short intervals.
- (3) *Forwards: assemble.* Each $\Omega_{1/n}$ has measure zero, and $D = \bigcup_n \Omega_{1/n}$ is a countable union of such sets, hence of measure zero.
- (4) *Backwards: split the partition in two.* On pieces meeting Ω_ε the oscillation may be as large as $2M$, but their total length is under ε by hypothesis. On the rest the oscillation is below ε , but the total length may be as large as $b - a$. Either way the product is small (Figure 12).
- (5) *Backwards: make the second kind finite.* Off the covering, every point has a neighbourhood of small oscillation; the leftover set is closed and bounded, so finitely many such neighbourhoods suffice and define a partition.
- (6) *Audit.* Ω_ε closed is used in step (5), to know the leftover set is compact. Countability is used once, in step (3). The bound $|f| \leq M$ is used once, in step (4) — which is why the theorem needs f bounded.

The backward half is drawn as a flow chart in Figure 11.

Proof sketch. Suppose f is integrable and fix n . Given $\eta > 0$, choose a partition with $U - L < \eta/n$. Each subinterval whose interior meets $\Omega_{1/n}$ contributes at least $(1/n)\Delta x_i$ to $U - L$, so such subintervals have total length below η ; they cover $\Omega_{1/n}$ apart from finitely many partition points, which are covered by intervals of arbitrarily small total length. Hence each $\Omega_{1/n}$ has measure zero and so does the countable union D .

Conversely, suppose D has measure zero and $|f| \leq M$. Fix $\varepsilon > 0$. The set Ω_ε is closed and bounded, hence compact, and of measure zero, so finitely many open intervals of total length less than ε cover it. Off their union every point has oscillation below ε , hence a neighbourhood on which f varies by less than ε ; compactness of the complement yields a finite subcover

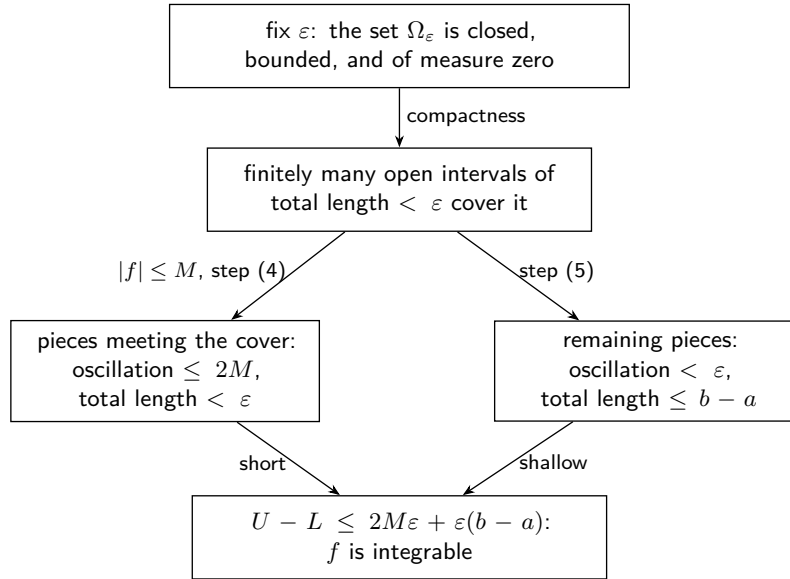


Figure 11: The backward half of Lebesgue's criterion (measure zero $\Rightarrow$ integrable) as a flow chart. Every subinterval of the partition is of one of two kinds, and each kind contributes a small product to $U - L$: the pieces meeting the cover are collectively short, the rest uniformly shallow. The forward half runs the same exchange rate in reverse, reading a small area as a small length (steps (2)–(3)).

and thus a partition. Splitting $U - L$ into the subintervals of the two kinds bounds it by $2M\varepsilon + \varepsilon(b - a)$. $\square$

The criterion settles the integrability row at a stroke.

function	discontinuous on	integrable
step function (E5)	a finite set	yes
Thomae (E6)	$\mathbb{Q}$, countable	yes
Dirichlet (E7)	every point	no
$x^2 \sin(1/x)$ (E3)	no point	yes

Two consequences deserve emphasis. First, E6 (Figure 13) is integrable while discontinuous on a *dense* set, so integrability is compatible with very bad pointwise behaviour and there is no arrow from integrability back to continuity. Second, E7 is bounded and not integrable, so boundedness alone is insufficient.

3.2 A derivative that is not integrable

Theorem 3.4 (Volterra). *There is a function F on $[0, 1]$, differentiable at every point, whose derivative is bounded but not Riemann integrable.*

How the construction runs

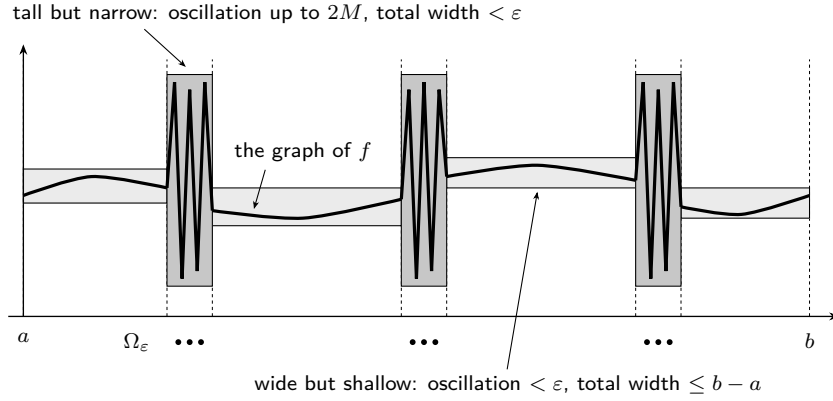


Figure 12: The two kinds of subinterval in the proof of Theorem 3.3. As in Figure 9, $U - L$ is the total area of the strips, one per subinterval, each just tall enough to contain its piece of the graph. Where the graph oscillates badly the strips are tall — up to $2M$ — but they stand exactly over the covered set Ω_ε , so their total width is below ε . Everywhere else the strips may be wide, but each is shallower than ε . Both kinds of area are therefore small, and $U - L \leq 2M\varepsilon + \varepsilon(b - a)$.

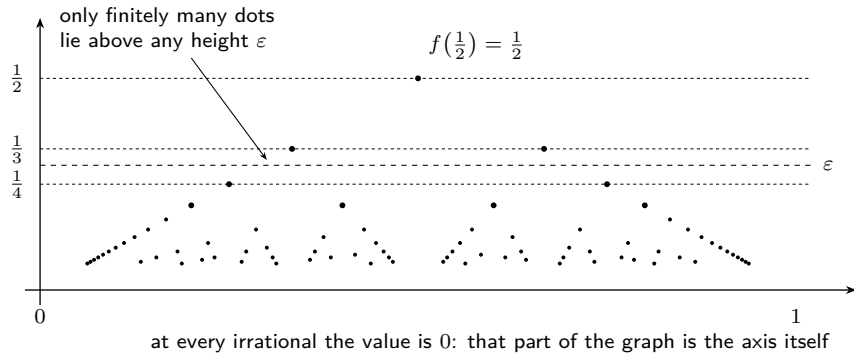


Figure 13: Thomae's function E6: the value at p/q in lowest terms is $1/q$, and 0 at every irrational. The dots on the level $y = 1/q$ are exactly the fractions of denominator q , so the levels descend as the dots on them multiply: points of large denominator are numerous and low. Above any fixed height ε (dashed) only finitely many dots remain — which is why f is continuous at every irrational, and why its discontinuities are only the rationals, a countable set. By Theorem 3.3 the function is integrable.

- (1) *Decide what is needed.* A derivative whose oscillation fails to vanish on a set of *positive* outer length — by Theorem 3.3, that alone makes it non-integrable.
- (2) *Build a bad set that is not null.* Run the middle-thirds construction of the Cantor set, but remove intervals whose lengths shrink fast enough that the remaining *fat Cantor set* K , though nowhere dense, fails to have measure zero.
- (3) *Plant oscillation in every gap.* In each removed interval place a scaled copy of E3, tapered so that both it and its derivative vanish at the endpoints; set $F = 0$ on K (Figure 15).
- (4) *Check differentiability on K .* The taper is what makes the difference quotient at a point of K tend to 0: each planted copy is dominated by the square of its distance to the gap's endpoints, so F is differentiable on K with $F' = 0$ there, and F' is bounded. Differentiability inside the gaps is inherited from E3.
- (5) *Collect.* Arbitrarily close to any point of K there are points where F' takes values near ± 1 , so $\Omega_1 \supset K$; and K does not have measure zero, so by Theorem 3.3 the derivative F' is not integrable.

Figure 14 draws the logic of the construction.

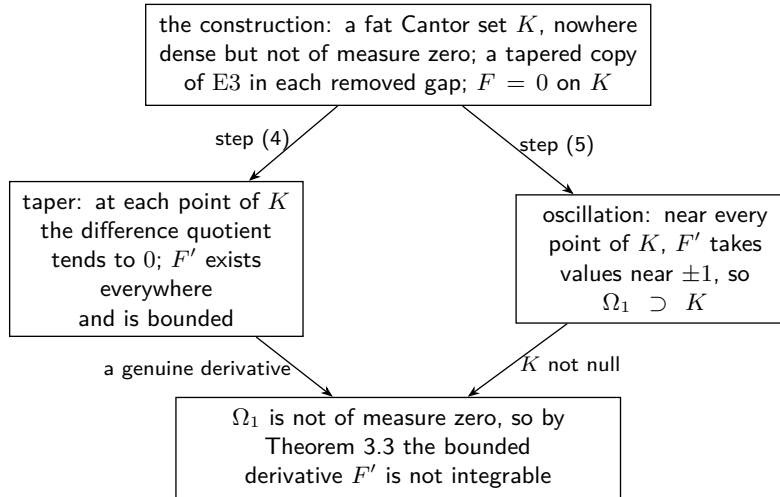


Figure 14: Volterra's construction as a flow chart. The two branches are independent: the taper makes F a genuine everywhere-differentiable function with bounded derivative, while the planted oscillation forces the discontinuity set of F' to contain K — and K was built not to be null, so Lebesgue's criterion refuses the integral. The step numbers refer to the walk above.

Remark 3.5. Theorem 3.4 is what forces the hypothesis in the Fundamental Theorem of Calculus. The statement “if $F' = f$ on $[a, b]$ then $\int_a^b f = F(b) - F(a)$ ” is false as it stands, because the integral on the left need not exist; one must assume f integrable. Together with Corollary 2.5 this shows the two properties are independent: E5 is integrable and is not a derivative, and Volterra's F' is a derivative and is not integrable.

Remark 3.6. For a bounded f on $[a, b]$ the four conditions — f continuous, f integrable, f possessing an antiderivative, and f both a derivative and integrable — are distinct. The first

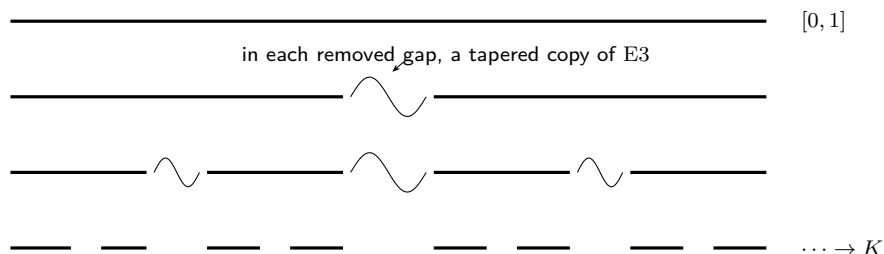


Figure 15: Volterra's construction. Removing middle intervals whose lengths shrink quickly leaves a fat Cantor set K — nowhere dense, yet not of measure zero. Planting a tapered copy of $E3$ in every removed gap produces a function differentiable everywhere whose derivative oscillates at every point of K , so that $\Omega_1 \supset K$.

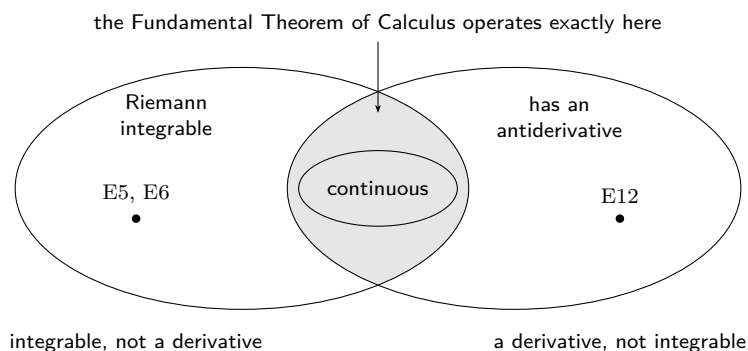


Figure 16: The four conditions of Remark 3.6 as a Venn diagram: each point of the picture is a single function on $[a, b]$, and each oval is a class of functions. The continuous functions are a small island inside the intersection, and the Fundamental Theorem of Calculus lives on the shaded lens: a function must be integrable *and* a derivative before $\int_a^b F' = F(b) - F(a)$ can even be stated. The two marked functions show that neither oval contains the other: the step function $E5$ and Thomae's function $E6$ are integrable but are not derivatives (Corollary 2.5), while Volterra's $E12$ is a derivative but is not integrable (Theorem 3.4).

implies the rest; the second and third are independent by the previous remark; and the fourth, their conjunction, is what the Fundamental Theorem requires (Figure 16).

4 Passing to the limit

Let $f_n \rightarrow f$ on a set E . Each property of the preceding sections may now be asked twice: of the f_n , and of f . Figure 17 tabulates the verdicts; the rest of the section proves them.

if every f_n has...	...does the limit f ?	
	pointwise conv.	uniform conv.
continuity	no (E9)	yes
integrability	no (E7)	yes
$\int_a^b f = \lim_n \int_a^b f_n$	no (E11)	yes
differentiability	no	no (E8, E10)

Figure 17: What survives a limit. Uniform convergence is strictly stronger than pointwise convergence, and it rescues the first three rows; the fourth it cannot rescue, because differentiation multiplies amplitude by frequency and uniformity bounds only amplitude. The repair for the last row is Theorem 4.4: assume uniform convergence of the derivatives f'_n instead.

Recall Definition 1.6: convergence is *pointwise* if $f_n(x) \rightarrow f(x)$ at each x , and *uniform* if $\sup_E |f_n - f| \rightarrow 0$, so that one index serves every point at once. Figure 18 shows the gap between the two on E9.

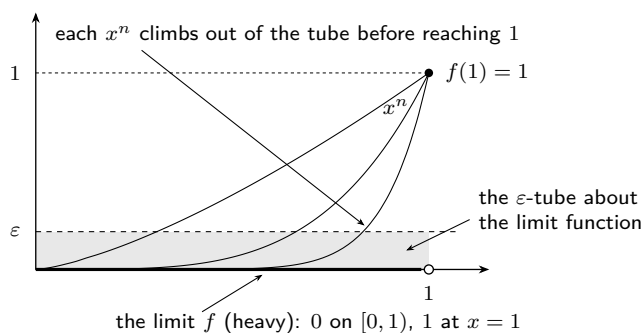


Figure 18: $f_n(x) = x^n$ on $[0, 1]$ (E9; three members drawn, later ones lower). The pointwise limit f — the heavy graph, 0 up to 1 and then the isolated value 1 — is discontinuous. Uniform convergence would require the whole graph of x^n , for large n , to stay inside the shaded ε -tube about f ; however large n is, the graph still climbs out before reaching 1, so the convergence is pointwise but not uniform.

Theorem 4.1. *If each f_n is continuous and $f_n \rightarrow f$ uniformly, then f is continuous.*

How the proof runs

- (1) *Locate the obstacle.* We must compare $f(x)$ with $f(x_0)$, and we know nothing about f directly — it is only a limit.

- (2) *Find a detour.* Each f_n is continuous, so route through one: from $f(x)$ up to $f_n(x)$, across to $f_n(x_0)$, and back down to $f(x_0)$ (Figure 19).
- (3) *Price the legs.* The two vertical legs are short by convergence; the horizontal leg is short by continuity of f_n . Three legs, so each is allotted $\varepsilon/3$.
- (4) *Check the order of choices.* This is where uniformity is spent. The index n must be fixed *before* x is, because the δ of the horizontal leg depends on n . Uniform convergence permits that order; pointwise convergence does not, since there the admissible n depends on the point.
- (5) *Audit.* Uniformity is used exactly twice, for the two vertical legs, and continuity of a single f_n once. No property of the index beyond its existence is used, which is why the same argument proves the corresponding statements for uniform limits of uniformly continuous or of bounded functions.

Proof. Fix x_0 and $\varepsilon > 0$. Choose n with $\sup |f_n - f| < \varepsilon/3$, then δ with $|f_n(x) - f_n(x_0)| < \varepsilon/3$ for $|x - x_0| < \delta$. For such x ,

$$|f(x) - f(x_0)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| < \varepsilon. \quad \square$$

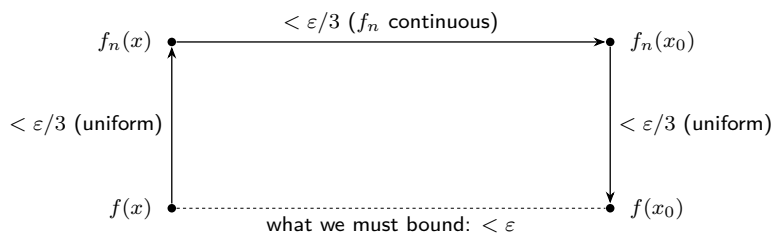


Figure 19: The three legs of the 3ε argument. The direct route along the bottom is unavailable, since nothing is known about f ; one detours through a single f_n . Uniform convergence is what permits that n to be chosen *before* x , which is exactly what pointwise convergence would not allow.

Uniformity is used for the outer two terms. Under pointwise convergence each can still be made small, but only after the point has been fixed, and the middle term then requires a δ depending on the index chosen for that point; the estimate does not close. E9 is what the failure looks like.

Theorem 4.2. *If each f_n is Riemann integrable on $[a, b]$ and $f_n \rightarrow f$ uniformly, then f is integrable and $\int_a^b f = \lim_n \int_a^b f_n$.*

Idea of the proof. Uniform convergence puts f inside a horizontal band of width $2\varepsilon_n$ about f_n . Upper and lower sums are monotone under such a comparison, so the strips of Figure 9 for f are no taller than those for f_n plus the band — and the band contributes at most $2\varepsilon_n(b - a)$ no matter how the partition is chosen. Two smallnesses are combined: one from n , one from the partition.

Proof. Put $\varepsilon_n = \sup_{[a,b]} |f_n - f| \rightarrow 0$, so that $f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n$. For any partition P ,

$$U(P, f) - L(P, f) \leq U(P, f_n) - L(P, f_n) + 2\varepsilon_n(b - a),$$

and choosing n and then P suitably makes the right side small, giving integrability. The same inequalities bound $|\int f - \int f_n|$ by $\varepsilon_n(b - a)$. $\square$

Theorem 4.3. *Uniform convergence does not preserve differentiability; and even when the limit is differentiable, $\lim_n f'_n$ need not equal f' .*

Proof. For the first clause, take the partial sums of E8: each is a finite sum of cosines, hence $\mathcal{C}^\infty$, and by the M -test they converge uniformly — to Weierstrass's function, which is differentiable at no point. So a uniform limit of smooth functions need not be differentiable anywhere.

For the second, let $f_n(x) = \sin(nx)/\sqrt{n}$ on $\mathbb{R}$ (E10; Figure 20). Then $\sup |f_n| = 1/\sqrt{n} \rightarrow 0$, so $f_n \rightarrow 0$ uniformly and the limit is smooth. But $f'_n(x) = \sqrt{n} \cos(nx)$ is unbounded and has no limit at any point; at $x = 0$, $f'_n(0) = \sqrt{n} \rightarrow \infty$ while $f' \equiv 0$. So even when the limit is differentiable, the derivatives may converge to nothing at all. $\square$

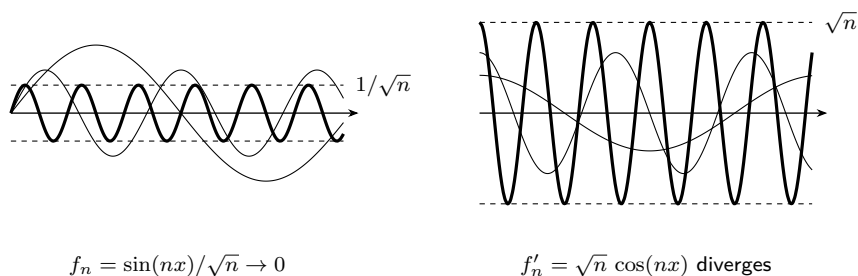


Figure 20: Why uniform convergence cannot control derivatives. The same three functions appear in both panels, and the heavy curve is the largest n of the three. On the left it is the *flattest*: its amplitude $1/\sqrt{n}$ (dashed) has collapsed furthest. On the right, after differentiating, the very same curve is the *steepest*: the factor n overwhelms the collapse, and its amplitude $\sqrt{n}$ (dashed) has grown furthest. Uniform smallness of a function says nothing about its slope: a wiggle may be short and steep at once.

Theorem 4.4 (Rudin 7.17). *Suppose each f_n is differentiable on $[a, b]$, the sequence $(f_n(x_0))$ converges for some $x_0 \in [a, b]$, and (f'_n) converges uniformly. Then (f_n) converges uniformly to some f , and $f' = \lim_n f'_n$.*

The hypothesis has moved onto the derivatives, and the functions themselves need only converge at one point. Differentiation is not continuous with respect to uniform convergence, so to differentiate a limit one must assume something about the derivatives; the same asymmetry is why term-by-term differentiation of a power series requires its own theorem while term-by-term integration does not.

Theorem 4.5. *Pointwise convergence preserves neither continuity, nor integrability, nor the value of the integral.*

Proof. Continuity fails by E9. For integrability, enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and let $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$; each f_n has finitely many discontinuities, hence is integrable by Theorem 3.3, while $f_n \rightarrow \mathbf{E7}$ pointwise and $\mathbf{E7}$ is not integrable. For the integral, the spikes $\mathbf{E11}$ (Figure 21) satisfy $f_n \rightarrow 0$ pointwise with $\int_0^1 f_n = 1$ for every n . $\square$

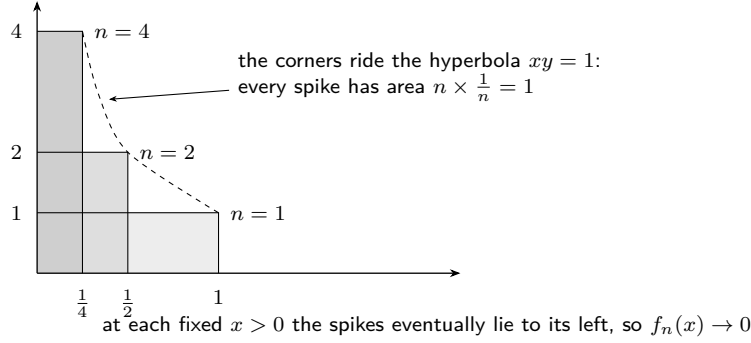


Figure 21: The spikes $f_n = n\mathbf{1}_{(0, 1/n)}$ ($\mathbf{E11}$), each enclosing area 1: as the base shrinks to $1/n$ the height grows to n , and the top-right corner $(1/n, n)$ rides up the hyperbola $xy = 1$. At each fixed $x > 0$ the spike eventually lies entirely to the left of x , so $f_n(x) \rightarrow 0$ pointwise; yet $\int f_n = 1$ for every n . The mass escapes upward, which no pointwise statement detects.

Remark 4.6. Pointwise limits are nonetheless not arbitrary. A pointwise limit of continuous functions is of Baire class one, and not every function is: the Dirichlet function $\mathbf{E7}$ is not a pointwise limit of continuous functions, by Baire's theorem [5, Lect. 28]. It does arise as a pointwise limit of the integrable functions in the proof above, but never of continuous ones.

Remark 4.7. The uniformity hypothesis in Theorem 4.2 is a limitation of the Riemann integral rather than of the statement. Under the Lebesgue integral the conclusion $\int \lim = \lim \int$ holds under a pointwise hypothesis together with an integrable dominating function — a hypothesis the spikes of $\mathbf{E11}$ conspicuously fail.

5 A gallery of counterexamples

Twelve functions carry the whole argument. Each entry gives the definition, the property the function has, the property it lacks, and the implication it refutes.

	function	refutes
E1	removable spike	limit $\Rightarrow$ continuous
E2	$ x $	continuous $\Rightarrow$ differentiable
E3	$x^2 \sin(1/x)$	differentiable $\Rightarrow \mathcal{C}^1$
E4	$\sin(1/x)$	bounded $\Rightarrow$ has a limit
E5	step function	integrable $\Rightarrow$ continuous
E6	Thomae	integrable $\Rightarrow$ disc. set nowhere dense
E7	Dirichlet	bounded $\Rightarrow$ integrable
E8	Weierstrass	continuous $\Rightarrow$ differentiable somewhere
E9	x^n on $[0, 1]$	pointwise preserves continuity
E10	$\sin(nx)/\sqrt{n}$	uniform conv. commutes with d/dx
E11	tall spikes	pointwise preserves $\int$
E12	Volterra	derivative $\Rightarrow$ integrable

E1 *Removable spike*. $f(x) = 0$ for $x \neq 0$ and $f(0) = 1$. The limit at 0 is 0 and the value is 1.

E2 *Absolute value*. $f(x) = |x|$ is continuous everywhere; at 0 the difference quotient $|h|/h$ equals ± 1 according to the side, so $f'(0)$ does not exist. Note f is Lipschitz, so that condition does not give differentiability either.

E3 $x^2 \sin(1/x)$. With $f(0) = 0$, the bound $|f(x)| \leq x^2$ gives $f'(0) = 0$, while for $x \neq 0$

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x},$$

which has no limit at 0. Thus f is differentiable everywhere with f' discontinuous at 0, the discontinuity being of the second kind as Corollary 2.5 requires. This function is the building block of E12.

E4 $\sin(1/x)$. Bounded by 1 with no limit at 0: along $x_n = 1/(2n\pi)$ the values are 0 and along $y_n = 1/(2n\pi + \pi/2)$ they are 1. Its oscillation at 0 is 2.

E5 *Step function*. $f = 0$ on $[0, 1)$ and $f = 1$ on $[1, 2]$. Integrable, having one discontinuity, and not continuous; by Corollary 2.5 it is also not a derivative.

E6 *Thomae's function*. $f(p/q) = 1/q$ in lowest terms and $f = 0$ at irrationals. Continuous at every irrational, discontinuous at every rational, and integrable with $\int_0^1 f = 0$ by Theorem 3.3, $\mathbb{Q}$ being countable. No function is continuous exactly at the rationals, so E6 has no mirror image [1, appendix to Chapter 4].

E7 *Dirichlet's function*. $\mathbf{1}_{\mathbb{Q}}$ is bounded with $\omega_f \equiv 1$, so it is nowhere continuous and not integrable; directly, every upper sum is 1 and every lower sum 0.

E8 *Weierstrass's function*. $f(x) = \sum_{k \geq 0} a^k \cos(b^k \pi x)$ with $0 < a < 1$, b an odd integer, and $ab > 1 + \frac{3}{2}\pi$. The series converges uniformly, so f is continuous by Theorem 4.1, and f is differentiable at no point. Its partial sums are $\mathcal{C}^\infty$ and converge uniformly, so E8 also witnesses that uniform convergence does not preserve differentiability (Theorem 4.3). By Baire's theorem the continuous functions differentiable somewhere are of first category in $\mathcal{C}[a, b]$, so this behaviour is typical rather than exceptional.

E9 x^n on $[0, 1]$. Each term is continuous and the pointwise limit is not; the convergence is not uniform, since $\sup_{[0, 1]} |f_n - f| = 1$ for every n .

E10 $\sin(nx)/\sqrt{n}$. Converges uniformly to 0 while the derivatives $\sqrt{n}\cos(nx)$ diverge everywhere, so the interchange $\lim f'_n = (\lim f_n)'$ fails under uniform convergence. Differentiation multiplies amplitude by frequency, and uniform convergence bounds only amplitude.

E11 *Tall spikes*. $f_n = n\mathbf{1}_{(0,1/n)}$, each integrable with $\int_0^1 f_n = 1$, tending pointwise to 0. The sequence admits no integrable dominating function — near 0 the envelope $\sup_n f_n$ grows like $1/x$ — which is exactly what dominated convergence would require.

E12 *Volterra's derivative*. The derivative F' of the function constructed in §3: bounded, a derivative by construction, and not Riemann integrable, because it oscillates on a nowhere dense set of positive measure. The function F itself is continuous, hence integrable; it is the derivative that witnesses the gap.

6 Concluding remarks

Three observations summarise the map.

The local chain is strict at every link and the witnesses are elementary: E1, E2, E3 suffice to separate having a limit, continuity, differentiability, and continuous differentiability.

Integrability is not part of that chain. It is global, it follows from compactness rather than from any pointwise hypothesis (Theorem 3.1), and it admits an exact characterisation in terms of the size of the discontinuity set (Theorem 3.3), for which oscillation is the natural language.

The interesting failures are all failures of interchange: of $\lim$ with $\int$, of $\lim$ with $\lim$, and of $\lim$ with d/dx . Uniform convergence repairs the first two and cannot repair the third, because differentiation amplifies precisely what uniformity controls. Each gap in the lattice invites the question of what hypothesis would close it: pointwise convergence strengthened to uniform restores continuity and the integral; uniform convergence transferred to the derivatives restores differentiation (Theorem 4.4); and replacing the Riemann integral by the Lebesgue integral restores the interchange under a domination hypothesis in place of a uniformity one.

References

- [1] *The Annotated Rudin*. A verbatim edition of [2] with commentary, figures, and per-chapter appendices drawing on [3, 4, 5, 6, 7]. The oscillation lemmas used in §3 are proved in the appendices to Chapters 2 and 4.
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